Finance 218 project

Wednesday, February 18, 2026 8:35 AM

Frontend

User chooses ticker and forecast horizon

☆ Use React.js ☆

Get Data

Use yfinance or Alpaca to get 1000 days of data (open, high, close, low, volume, date)

Compute log returns and store as pandas dataframe for later use

w/ Numpy or Pandas

Compute mean, stdev, variance, skewness, kurtosis

☆ Global values ☆

Compute technical indicators

Use pandas-ta or manually

Trend direction

- 50 and 200-day SMA
  - $\frac{SMA50 - SMA200}{SMA200}$
- MACD (12, 26, 9)

Momentum

- 14-day RSI

Volatility

- Bollinger bandwidth (20 day, 2σ)
  - $\frac{\text{upper} - \text{lower}}{\text{middle}}$

Volume

On Balance Volume

OBV direction vs price direction

SMA 200
- MACD (12, 26, 9)

Upper-Lower
Middle
- ATR (14 day)
- Hist. Volatility Ratio
$\frac{10 \text{ day } \sigma}{60 \text{ day } \sigma}$

price direction

☆ Computed for every day possible

Identify regimes based on TAs w/ python conditionals

Low-Vol Bull
SMA 50 > SMA200
BBW < median
HVR < 1.0
OBV confirming price trend

High-Vol Bull
SMA 50 > SMA200
BBW > median or
HVR > 1.2
RSI > 65

High-Vol Bear
SMA50 < SMA200
BBW > median
HVR > 1.2
RSI < 35
OBV diverging from price

Sideways/Consolidation
Tight SMA spread
RSI 40-60
BBW ≤ median
MACD hist oscillating near zero

Compute Regime specific parameters (cumulative)

μ - mean daily log returns

σ - stdev of log returns

GARCH(1,1) - (ω, α, β)

t-test w/ t degrees of Freedom

Jump parameters / regime
λ - jump intensity

Jump parameters / regime

λ - jump intensity

μ_J - Mean Jump size

σ_J - jump size stder

minimum day filter

Regime must last for at least 10 days

Calculate transition matrix

Count day-to-day regime transitions

Example $P(\text{stay}) = 90\%$

$P(\text{HV Bull} \to \text{Sideways}) = 1\%$

Start C++ Monte Carlo Simulations

use pybind11

Send:

- Current Price (S_0)
- Current Regime
- Transition Matrix
- per-regime sets
- Sim config

10,000 paths

forecast horizon

$dt = 1/252$

Simulation

Analyze Simulation Results

use python pandas, numpy, scipy

Analyze simulation results

use python pandas, numpy, scipy

- Conf. intervals @ each time step
  50% CI, 95% CI
- Prob. Distributions
- Risk metrics
- Current Regime context

SHOW ON FRONTEND

wow! □□□□□□□